

CS 636 Semester 2024–2025-II: Assignment 1

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NOTE:

All the codes for this assignment have been compiled and executed on KD first floor workstation with ID: **CSEWS33**.

Solution

The details of implementations of both algorithms have been commented alongside the code.

Steps to compile and run the code

- Comment the code related to timings before proceeding to generate the data races.

Compilation Command: `g++ main.cpp -o main`

Run Command: `./main <djit | fasttrack> <file_path>`

Example: `./main djit hbanalysis.log`

Steps to generate readings

1. Comment the call to `printDataRaces()` function in `main.cpp`.
2. Uncomment the lines of code related to timings and re-compile the code.
3. In the `timer.sh` file, supply the name of the trace file and run it.
4. Pair of 5 readings will be generated one each for DJIT⁺ and FastTrack.

Results

The following tables provide the timings of DJIT⁺ and FastTrack algorithms for three different trace files over **5** runs.

Table 1. hbanalysis.log

Iteration	DJIT+ (in sec)	FastTrack (in sec)
1	46.096	45.97
2	46.72	45.75
3	46.26	46.19
4	46.30	45.87
5	46.20	45.94

Output files (for each algorithm) sorted by the memory locations and screenshot for the timing output of 5 iterations have been attached.